TEXORA · LECTURE SHEET

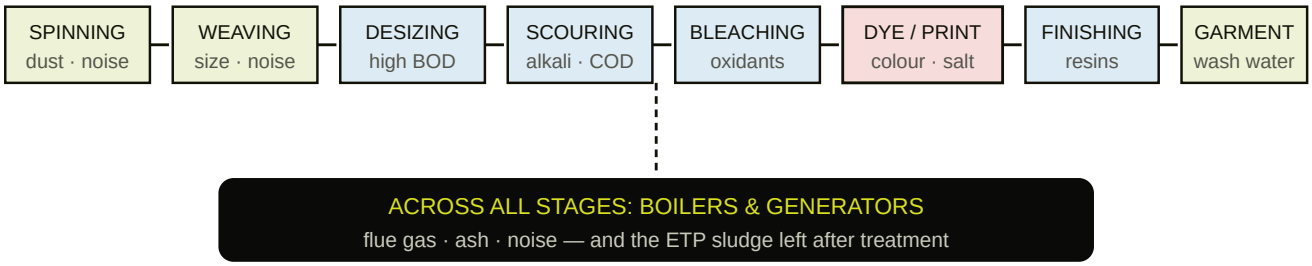
Types of Textile Pollution and Their Control/Treatment

How each stage of the textile chain generates water, air, noise, solid-waste and chemical pollution — and the source-control and treatment measures matched to each type.

TE-4204 · ENVIRONMENTAL ENGINEERING AND POLLUTION CONTROL · TOPIC 3 · LEVEL 4 TERM 2

1. Why the textile chain pollutes at every stage

Textile manufacturing is a long chain of mechanical, thermal and wet-chemical operations. The mechanical stages (spinning, weaving, knitting) mainly produce fibre dust, noise and solid waste. The **wet-processing stages** — desizing, scouring, bleaching, dyeing, printing and finishing — dominate the pollution load because they use large volumes of water and a wide chemical inventory, and discharge most of what they use. A pollution survey of a factory therefore walks the chain stage by stage, asking at each one: what enters, what leaves in the product, and what leaves as waste?



Dyeing/printing (highlighted) is usually the single largest source of colour, salt and toxicity in the effluent.

Figure 1 — Pollution along the textile chain. Dry stages give dust, noise and solid waste; wet stages give effluent; energy plant serves every stage and adds air emissions and ash.

2. Water pollution — the dominant stream

PROCESS	CHARACTERISTIC POLLUTANTS	WHY IT MATTERS
Desizing	Starch or synthetic size — very high BOD	Often the largest single share of the effluent's oxygen demand

Scouring	Caustic soda, waxes, pectins — high pH, high COD	Alkaline, hot, grease-bearing stream
Bleaching	Residual peroxide or chlorine compounds, high pH	Oxidising stream; chlorinated by-products if hypochlorite is used
Mercerising	Strong caustic soda (partly recoverable)	High alkalinity unless caustic recovery is installed
Dyeing	Unfixed colour, salt (often tens of grams per litre), alkali, metals in some dye classes	Colour is visible at very low concentration; salt (TDS) is hard to remove
Printing	Pigments, binders, urea, thickener wash-off	Colour and nitrogen load from screen/roller washing
Finishing	Resins and cross-linkers (formaldehyde-releasing), softeners, coating polymers	Toxicity and COD from small but concentrated streams
Garment washing	Pumice solids, enzymes, detergents, indigo	High suspended solids and colour

Control follows a strict order of preference. **At source:** substitute lower-impact chemicals (peroxide for hypochlorite, low-salt reactive dyes, enzymes for harsh desizing chemicals), dose accurately, recover what can be recovered (caustic from mercerising, heat from hot effluent, reusable rinse water by counter-current flow), and keep concentrated small streams (dye-bath residues, print-paste washings) separate for targeted treatment instead of diluting them into the whole flow. **End of pipe:** the combined effluent passes to an effluent treatment plant (ETP) — screening, equalisation, neutralisation, chemical coagulation for colour and solids, biological oxidation for BOD, and clarification — designed in detail in Topic 6.

WORKED EXAMPLE — UNFIXED DYE AND SALT REACHING THE DRAIN

A dyehouse processes 1,000 kg of cotton fabric per batch with reactive dye at 2 % on weight of fabric, at a liquor ratio of 1:8. Dye exhaustion is 70 %.

Dye applied = 1,000 × 0.02 = **20 kg**. Fixed on fibre = 70 % × 20 = 14 kg.

Dye discharged with the spent bath = 20 − 14 = **6 kg per batch** — before counting further losses by hydrolysis and wash-off.

Bath volume = 1,000 × 8 = 8,000 L. If the recipe also uses salt at 60 g/L, salt discharged = 60 × 8,000 = 480,000 g = **480 kg per batch**.

Lesson: colour control starts with exhaustion and fixation (right dye, right recipe, right-first-time dyeing); salt, once dissolved, has no cheap removal route — only using less of it works.

3. Air pollution

Air emissions come from **energy plant** (boilers and generators: particulates, SO_2, NO_x, CO — treated in depth in Topic 4), from **process operations** (fibre and fabric dust in spinning and raising; singeing fumes; stenter exhaust carrying vaporised oils and finishing chemicals; solvent vapours from coating and dry cleaning), and from **storage and handling** (chemical vapours, dust from powdered dyes and auxiliaries). Effects range from worker exposure inside the shed to odour complaints and deposition in the neighbourhood. Controls: enclose and extract at the point of generation with local exhaust ventilation; clean extracted air (filters for dust, scrubbers or after-burners for fume-laden stenter exhaust); enclose dye-powder handling and use dust-free (grain/liquid) dye forms; and maintain burners for clean combustion.

4. Noise pollution

Weaving sheds (loom impacts), spinning halls, compressors, generators and pumps generate continuous noise, commonly in ranges that damage hearing over a shift (Topic 5 gives the measurement and control method). Controls: machine maintenance and balancing, anti-vibration mounts, acoustic enclosures for generators and compressors, sound-absorbing shed surfaces, layout separation of noisy plant, and hearing protection with audiometric monitoring for exposed workers.

5. Solid-waste pollution

Solid streams include **ETP sludge** (the pollutant mass removed from effluent reappears here — treatment transfers pollution between media, it does not annihilate it), fabric and yarn cutting waste, fibre fly and floor sweepings, packaging (polythene, cartons, cones), empty chemical containers (a hazardous stream if residues remain), boiler ash, and general refuse. Controls follow the waste hierarchy — prevent, reduce, reuse, recycle, recover, dispose: cutting waste is minimised by marker efficiency and sold for recycling (shoddy, wiping cloths); containers are triple-rinsed and returned or handled as hazardous waste; sludge is dewatered and sent to approved disposal or controlled uses (Topic 7); ash goes to brick-making or land fill under conditions that prevent leaching.

6. Chemical and thermal pollution; occupational exposure

Two further types cut across the others. **Chemical pollution** in its specific sense means toxic substances — formaldehyde from resins, banned amine-releasing azo dyes, heavy metals, alkylphenol ethoxylates, chlorinated carriers — whose harm is toxicity rather than load; control is by **substitution and inventory discipline** (restricted-substance lists, supplier declarations, chemical inventory with safety data sheets) because end-of-pipe treatment handles them poorly. **Thermal pollution** — hot

effluent and cooling water — lowers dissolved oxygen in receiving water and stresses aquatic life; control is heat recovery (which also saves fuel) and cooling before discharge. Inside the factory the same streams are **occupational exposures** — the worker at the machine is the nearest receptor — so occupational hygiene (ventilation, protective equipment, exposure monitoring) and environmental control are one programme, not two.

POLLUTION TYPE	MAIN TEXTILE SOURCES	FIRST-LINE CONTROL	TREATMENT / FINAL ROUTE
Water	Wet processing, garment washing	Substitution, recovery, right-first-time processing, segregation	ETP (physical–chemical–biological)
Air	Boilers, stenters, dust points	Clean combustion, enclosure + extraction	Cyclone / scrubber / ESP / bag filter; after-burning
Noise	Looms, generators, compressors	Maintenance, mounts, enclosures	Barriers, absorption, hearing protection
Solid waste	Sludge, cuttings, packaging, ash	Waste hierarchy, marker efficiency	Recycling, controlled use, secure disposal
Chemical (toxic)	Dyes, resins, auxiliaries	Substitution, RSL/MRSL discipline, inventory control	Segregated handling; no dilution solution
Thermal	Hot effluent, cooling water	Heat recovery	Cooling before discharge

Glossary

Liquor ratio — mass of treatment liquor per unit mass of textile material (e.g. 1:8). **Exhaustion** — percentage of applied dye taken up by the fibre from the bath. **Fixation** — percentage of applied dye permanently bonded to the fibre. **Right-first-time dyeing** — achieving the correct shade in one pass, avoiding reprocessing and its extra water, chemicals and effluent. **ETP** — effluent treatment plant. **TDS** — total dissolved solids, mainly salts in textile effluent. **RSL / MRSL** — restricted-substances list (for products) / manufacturing restricted-substances list (for process inputs). **Local exhaust ventilation** — extraction of contaminated air at its point of generation. **Sludge** — the solid residue separated from wastewater during treatment.

Key points

• Walk the chain: dry stages → dust, noise, solid waste; wet stages → effluent; energy plant → air emissions, ash and noise everywhere.

• Dyeing/printing usually dominates colour, salt and toxicity; desizing often dominates BOD.
• Control order: substitute and recover at source first, segregate concentrated streams, treat the combined flow last.
• In the worked example, 30 % unfixed dye means 6 kg of dye per 1,000 kg batch reaches the drain, with 480 kg of dissolved salt per 8,000 L bath that no cheap treatment removes.
• Treatment transfers pollution between media (effluent → sludge); toxic chemicals are controlled by substitution, not dilution.

Practice

1. For each of desizing, dyeing and finishing, name the characteristic pollutants and one source-control measure.
2. A batch of 2,000 kg fabric is dyed at 3 % owf with 75 % total fixation. How much dye reaches the drain? At a liquor ratio of 1:10 and salt dose 50 g/L, how much salt is discharged per batch?
3. Explain why segregating a small, concentrated print-paste washing stream is better than mixing it with the full effluent flow.
4. Give one example each of how air, noise and solid-waste pollution from a composite factory is controlled at source rather than after generation.
5. Why is sludge management part of water-pollution control, not a separate subject?

Answer key

1. Desizing: starch/synthetic size, very high BOD — enzymatic desizing and size recovery/reuse where practicable. Dyeing: colour, salt, alkali — high-fixation/low-salt dyes and right-first-time dyeing (also acceptable: accurate automatic dosing). Finishing: formaldehyde-releasing resins, softeners — low-/zero-formaldehyde resin substitution.
2. Dye applied = 2,000 × 0.03 = 60 kg; unfixed = 25 % × 60 = **15 kg to drain**. Bath volume = 2,000 × 10 = 20,000 L; salt = 50 × 20,000 = 1,000,000 g = **1,000 kg per batch**.
3. The pollutant mass is the same either way, but in a small segregated stream it is concentrated: it can be coagulated, oxidised or hauled for treatment at low volume cost, and it cannot shock-load or colour the entire ETP flow. Dilution spreads the same mass through thousands of cubic metres, where removal is slower, costlier per unit removed and sometimes impracticable.
4. Air: burner tuning/maintenance for complete combustion (or enclosing a dust point with extraction). Noise: anti-vibration mounts and machine balancing (or acoustic enclosure of a generator). Solid waste: marker planning to reduce cutting waste at source (any one valid example each).
5. Because the pollutant mass removed from water is concentrated into sludge; if the sludge is dumped where it leaches or washes back into water, the pollution returns. Water treatment is only complete

when the resulting sludge has a safe final route.